

a Shared evaluation embedding

3,964 cells across eight held-out species



● *Arabidopsis thaliana* ● *Gossypium bickii*
● *Fragaria vesca* ● *Gossypium hirsutum*
● *Brassica rapa* ● *Triticum aestivum*
● *Catharanthus roseus* ● *Eutrema salsugineum*

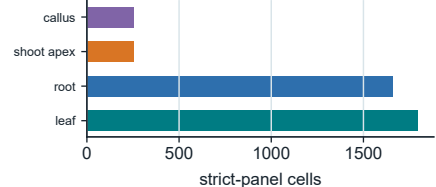
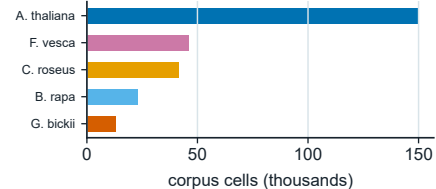
b Ontology states in the same coordinates

State structure is separated from species identity

● other ● xylem ● cortex ● meristem or stem cell niche ● epidermis ● cell cycle state ● vascular stele ● phloem ● endodermis ● root cap

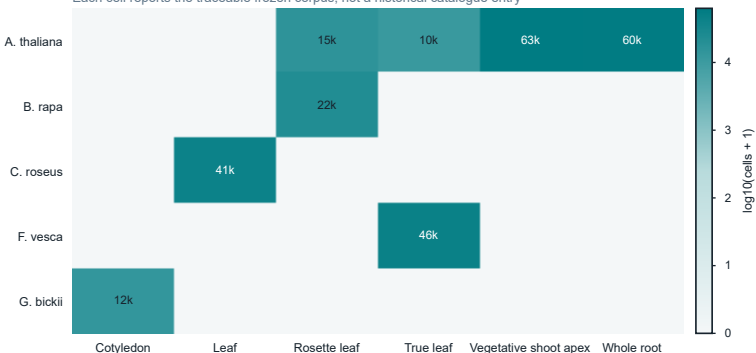
c Frozen corpus profile

272,132 cells, 209,406 genes



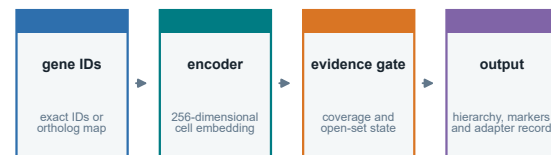
d Training data are distributed across species and organs

Each cell reports the traceable frozen corpus, not a historical catalogue entry



e A protocol-aware annotation contract

The model records assumptions that change the meaning of an annotation result



The frozen profile, inner-fold choice, target support and deployment head are never presented as one metric.